

Preceding Emptiness: Alternative Arabic Typographic Technologies

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1 DESCRIPTION OF ARTWORK

Throughout its history, the formal evolution of written language has been driven by the technology used to render that language—from cuneiform to stone-carved Roman capitals, and up to letterpress and pixels. This phenomenon has accelerated rapidly in the last century with typography adapting to new display possibilities: such as segmented LED displays in early electronics, to more contemporary storage and rendering standards, such as postscript. The locality of these innovations has been primarily situated in the West, where the Latin typographic script is the predominant form of written language. The opportunities presented by these technologies generated pressure for alternative scripts, such as written Arabic, to conform to the formal constraints of these Latin-based technologies, which, in some cases, have undermined the traditional formal structures of the Arabic script.

Preceding Emptiness is the latest light installation in a body of work imagining alternative typographic histories by constructing unconventional language display technologies. It imagines a timeline where Arabic played a leading role in the development of typographic technology. It explores these themes by designing an Arabic typeface within a modular segmented grid, made of 60cm LED tubes—a standard unit for lighting. This constraint is reminiscent of the heritage and development of Arabic calligraphy, where the tools and material units lead to the historical innovations of the script's forms. The square Kufic inscriptions in historic Islamic architecture are a prime example—by designing the script to be embedded within the structure of the wall of bricks, they developed



a new form for the Kufic script. Finding elements from traditional practices, and reimagining them within contemporary technological structures, allows discoveries to emerge. The grid iterations were a process to find a striking balance between legibility and ingenuity, where the core of the letters was preserved while pushing the forms to new boundaries. These speculative typographic artifacts aim to open new discourse and formal potentials for examining the current state of Arabic typography and calligraphy, and its future directions. The hexagonal segmented display is a custom-built, modular, computational installation. Each LED is individually switchable, controlled from a central hub that supplies power and routes communication. The system is custom-designed and fabricated from the ground up. The bright, bold, blinking segments cycle to display a series of Arabic words sourced from an archive of conference proceedings of the Arab School of Science and Technology. The words are Arabized terms of code statements for Arabic software tools that were first developed by Arab scientists and researchers in the 1980s—highlighting the presence of Arabic in early technology research. The words drive toward an unwritten landscape populated by unconventional technologies, that in turn shape new typographic structures—a new symbiosis between technology, typography, and tradition.

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2 PROJECT URL

<https://xlab.qatar.vcu.edu/Preceding-Emptiness>